Bioinformatics Data Analysis

The Unix Command Line

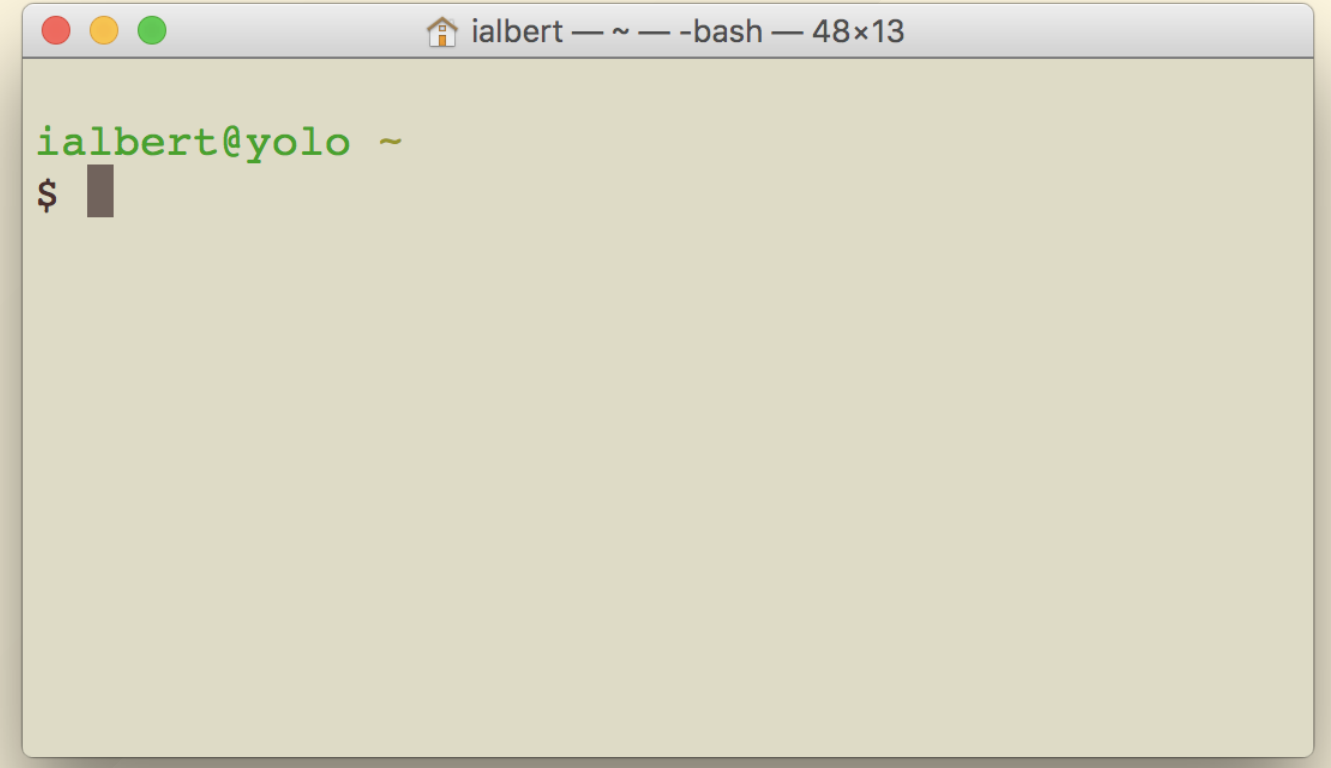
How does the command line work?

1. You type in "action" words that the computer executes.
2. You may combine these words to form longer commands.
3. You may redirect the results of commands into files

These are universally useful skills - applicable to any domain of science.

Your "terminal" or "shell"

A little window into the "guts" of your computer



What is the "terminal"

A terminal is a program that does two main things.

1. It allows you to type input into the computer: run programs, move/view files
2. It allows you to see the output from those programs.

All Unix machines will have a terminal program available.

We will show commands typed into or printed by the terminal: `like this`

What is a filesystem?

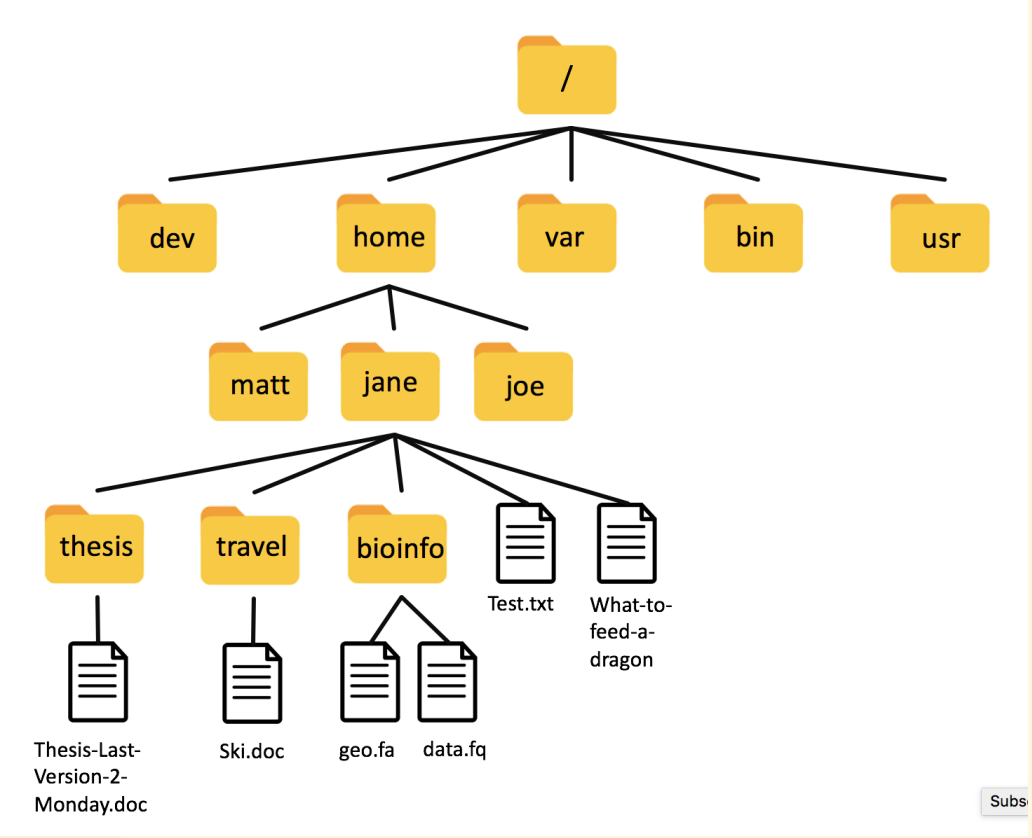
The "filesystem" is the term that describes how data is arranged on a computer.

The filesystem is made out of *directories* and *files*.

Windows started calling directories as *folders* - it is a synonym for *directory*.

A directory may contain files and other directories.

A filesystem example



Your terminal will start in **your home** directory.
Above it may be the `jane`, `matt` or `joe` based on login.

How to tell where you are

Print your current directory

```
pwd
```

for me it prints:

```
/Users/ialbert
```

What does it print for you?

Absolute paths

The location can be specified via a so-called "path". Nodes of the tree are separated via the `/` character.

```
/home/jane/travel/Ski.doc
```

The above is an *absolute path* to the file.

Relative paths

If the path below is a valid absolute path

```
/home/jane/travel/Ski.doc
```

Then if you are in the `home/jane` directory, then you could access the file with

```
travel/Ski.doc
```

There is no `/` at the start. This is a *relative* path.

How do I get help on a command?

Type:

```
man
```

Output:

```
What manual page do you want?
```

Now try:

```
man ls
```

Press `q` to exit.

How do I know what commands I could be using?

Learn them from examples. Some are "logical" shorthand notations, others not so much.

- `ls` - list files
- `mv` - move (rename) files
- `rm` - remove files
- `cp` - copy files

A few more essential commands

- `cd` - change directory
- `pwd` - print working directory
- `mkdir` - make directory
- `rmdir` - remove directory

  mkdir foo
  cd foo
  pwd
  mkdir bar
  cd
  pwd

Where are you now? Why? How do you get to `bar`?

What kind of commands are there?

These usually come with your computer:

1. Shell commands: `cd`, `pwd`, ...
2. Operating system commands: `cp`, `mv` ...

These you will have to install yourself:

3. Domain-specific specialized programs: `bwa`, `edirect`

What are command "flags"?

A flag is a small *decoration* used to customize the output of the tool. Compare the output of:

```
ls
```

to:

```
ls -l -h
```

The "flags" are `-l` and `-h`.

Flags usually may be combined into one as `-lh`

Everyone makes errors all the time

Using the command line can be error-prone - a simple typo makes things crash - sometimes spectacularly but mostly pitifully.

We all make errors all the time - that is normal.

The difference between a novice and someone skilled is in the efficiency of their error recovery.

You'll get used to immediately recognizing and recovering from 95% of the errors.

What causes this error?

Suppose you type

```
bwrapper
```

and you get the output:

```
-bash: bwrapper: command not found
```

What caused this error? How do you solve it?

This takes care of a large number of your initial errors.

What causes this other error?

Suppose you type

```
cat work/ranktest.txt
```

and you get the output:

```
cat: work/ranktest.txt: No such file or directory
```

What caused this error? How do you solve it?

Ok, another big chunk of your errors is now solved.

Do you know where you are? Do you know what is there?

Any time you run you get stuck make sure you know the answer the two questions in the title. You can find out those with

```
pwd
```

then

```
ls
```

You'll be surprised how often that helps you diagnose a problem.

Where to practice more?

This course is not a full introduction to the command line.

Use online command line tutorials to develop these skills further.

Check the links for this lecture for further details.